Resistance of a conductor is said to be one ohm if a current of one ampere flows through it when a potential difference of one volt is maintained across its ends.

$$1 \text{ ohm} = \frac{1 \text{ volt}}{1 \text{ ampere}}$$

Solved Problem-3

Calculate the resistance of a conductor through which a current of 2 A passes, when the potential difference between its ends is 30 V.

Solution:

Current through the conductor I = 2 A,
Potential Difference V = 30 V

From Ohm's Law: $R = \frac{V}{I}$.

Therefore, $R = \frac{30}{2} = 15\ \Omega$

4.6 ELECTRICAL RESISTIVITY & ELECTRICAL CONDUCTIVITY

4.6.1 Electrical Resistivity

You can verify by doing an experiment that the resistance of any conductor 'R' is directly proportional to the length of the conductor 'L' and is inversely proportional to its area of cross section 'A'.

$R \propto L$, $R \propto \frac{1}{A}$,

Hence , $R \propto \frac{L}{A}$

Therefore, $R = \rho \frac{L}{A}$ (4.4)

Where, ρ (rho) is a constant, called as electrical resistivity or specific resistance of the material of the conductor.

From equation (4.4), $\rho = \frac{RA}{L}$

If L = 1 m, A = 1 m^2 then, from the above equation $\rho = R$

Hence, the electrical resistivity of a material is ***defined as the resistance of a conductor of unit length and unit area of cross section***. Its unit is **ohm metre**.

Electrical resistivity of a conductor is a measure of the resisting power of a specified material to the passage of an electric current. It is a constant for a given material.



Nichrome is a conductor with highest resistivity equal to $1.5 \times 10^{-6}\ \Omega$ m. Hence, it is used in making heating elements.

4.6.2 Conductance and Conductivity

Conductance of a material is the property of a material to aid the flow of charges and hence, the passage of current in it. The conductance of a material is mathematically ***defined as the reciprocal of its resistance*** (R). Hence, the conductance '**G**' of a conductor is given by

$$G = \frac{1}{R} \qquad (4.5)$$

Its unit is ohm^{-1}. It is also represented as 'mho'.

The reciprocal of electrical resistivity of a material is called its electrical conductivity.

$$\sigma = \frac{1}{\rho} \qquad (4.6)$$

Its unit is ohm^{-1} $metre^{-1}$. It is also represented as mho $metre^{-1}$. The conductivity is a constant for a given material. Electrical conductivity of a conductor is a measure of its ability to pass the current through it. Some materials are good conductors of electric current. Example: copper, aluminium, etc. While some other materials are non-conductors of electric current (insulators). Example: glass, wood, rubber, etc.

Conductivity is more for conductors than for insulators. But, the resistivity is less for

10th_Science_Unit-4.indd 46 06-12-2021 18:29:11